

Greedy Algorithms work step-by-step, and always choose the steps which provide immediate profit/benefit. The choice made by the greedy approach does not consider the future consequences.

The greedy technique is used for optimization problems.

Greedy algorithms may not always lead to the optimal global solution, because it does not consider the entire data.

In some cases, making a decision that looks right at that moment gives the best solution (Greedy), but in other cases, it doesn't.

Greedy Choice Property

This property says that the globally optimal solution can be obtained by making a locally optimal solution (Greedy). The choice made by a Greedy algorithm may depend on earlier choices but not on the future. It iteratively makes one Greedy choice after another and reduces the given problem to a smaller one.

The selected choice must possess the following characteristics:

1. Feasible – It must fulfil local constraints.
2. Optimal – It must be the best at that stage.
3. Irrevocable – It cannot be changed once it is made.

Optimal Substructure

A problem exhibits optimal substructure if an optimal solution to the problem contains optimal solutions of the subproblems. That means we can solve subproblems and build up the solutions to solve larger problems.

All greedy algorithms follow a basic structure:

1. Declare an empty result.
2. Make a greedy choice to select and check if the choice is feasible. If it is feasible, add it to the result.
3. Return the result.